

# Subham Bhattacharya

Backend Engineer · Python

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Backend engineer focused on Python, with merged contributions to three widely used Python open-source projects and project experience in multi-tenant data isolation, role-based access control and asynchronous I/O under strict type checking. Computer Science undergraduate at BITS Pilani (online), available to start full-time within 7–10 days.

## TECHNICAL SKILLS

**Languages** Python, TypeScript, SQL, C, HTML/CSS

**Backend** FastAPI, Flask, ASGI/WSGI, REST API design, PostgreSQL, SQLite, relational data modelling, multi-tenant architecture, session authentication, RBAC, CSRF protection, password hashing, HTTP semantics, Jinja2

**Async & Concurrency** `asyncio`, `httpx`, bounded concurrency, per-host rate limiting, retries and backoff, timeouts, `robots.txt` compliance

**Typing, Testing & Quality** `mypy --strict`, Protocols, generics, dataclasses, generators, decorators, context managers, iterators, `pytest`, regression testing, `ruff`, ESLint, pre-commit hooks, GitHub Actions, CI, code review

**Infrastructure & Web** Git, Linux, Bash, uv, Cloudflare Pages/Workers, GraphQL, Content-Security-Policy, Next.js, React, Tailwind CSS, Zod

## OPEN SOURCE CONTRIBUTIONS

**OpenTelemetry** — CNCF observability framework · Prometheus remote-write exporter

[PR #5134](#) — merged

- Closed a silent TLS verification bypass: `insecure_skip_verify: False` was turning certificate checks *off*, exposing metric exports to man-in-the-middle interception. Negated the flag to match Prometheus `remote_write` semantics, with a parametrized regression test (fails before, passes after) and an end-to-end check against a self-signed HTTPS server; approved by two maintainers.

**Litestar** — high-performance Python ASGI web framework

[PR #4990](#) — merged

- Corrected HTTP semantics for multipart form part-limit violations, returning `413 Request Entity Too Large` in place of `400`; shipped with regression tests, a changelog entry and a breaking-change note. Reviewed and merged by a core maintainer.

**Strawberry GraphQL** — GraphQL library for Python

[PR #4591](#) · [#4581](#) — merged

- Corrected Strawberry's `dataclass_transform` metadata (PEP 681) to declare `order_default=False`, aligning static type-checker behaviour with runtime dataclass semantics — the previous value made `mypy` and `ty` reject valid user-defined comparison methods. Applied across related decorators with regression coverage; separately updated resolver docs to be runnable as written.

## PROJECTS

**Postmark** — AI-Powered Email Threat Detection Platform

[Demo](#)

Python · FastAPI · `async services` · PostgreSQL (Smart India Hackathon 2026 — selected)

- Team lead** of six engineers; selected to build SIH26106 — email threat detection with geolocation and forensic intelligence. Owned backend architecture and split delivery across backend, DevOps, ML and frontend.
- Built the FastAPI backend (`async handlers`, DB schema, data-access layer) and the team's test and git workflows; repo private by design.

**site-auditor** — Concurrent Site Crawler & Link-Health Auditor

[Code](#)

Python 3.12 · `asyncio` · `httpx` · `pytest` · `mypy --strict`

- Async fetch layer: pooled `httpx.AsyncClient` under explicit limits, per-host request spacing, `robots.txt` parsed and honoured. Retry uses full jitter (uniform over the whole backoff window, not a fraction), so concurrent failures don't resynchronise into a thundering herd.
- Frontier acts as the crawl's gatekeeper: URL normalisation and deduplication, depth and page caps, host scoping, and the seed host rebound after redirects so a redirecting seed does not silently reject every link it yields.
- Hand-written fakes satisfy the `Fetcher` protocol structurally, not by inheritance, so the retry path runs against a real failing-then-succeeding fetcher with no mocking library in the suite; `mypy --strict` and `ruff` green from the first commit.

**OrderFlow** — Multi-Tenant B2B Order Management

[Code](#) · [Demo](#)

Python · Flask · SQLite · Jinja2 · GitHub Actions

- Shared schema with every query scoped by `company_id`, so no tenant reads or writes across an organisational boundary; role separation via `admin_required` / `client_required` decorators, server-side sessions, password hashing and CSRF protection.
- Modelled the order lifecycle as an explicit state machine (pending → approved → fulfilled / rejected), validating each transition server-side rather than trusting client state; currency held as integer paise rather than floating point, eliminating rounding drift across totals and invoices.

## PROFESSIONAL EXPERIENCE

**Managing Director**

May 2024 — Present

Agam Agrovet Private Limited

Kolkata, West Bengal

- End-to-end ownership of an animal nutrition manufacturer (production scheduling, quality control, 40+ suppliers, 60+ customers); digitised purchase-order tracking and automated payables/receivables, cutting procurement turnaround and overdue payments by 40% each.

**Managing Director**

Sep 2022 — Present

Anandam Breeding Private Limited

Kolkata, West Bengal

- Lead day-to-day operations, supply chain, inventory control and distribution for an animal health products business, managing 20 people.

## EDUCATION & CERTIFICATIONS

BSc (Hons), Computer Science, BITS Pilani (online) · Semester 1 GPA 10.00/10.00 ([grade sheet](#)) · Mar 2026 — Mar 2030

**Certifications** CS50x, Harvard (CS50 AI in progress) · Python for Everybody (4 of 5), Michigan · LFC102 Open Source Community, Linux Foundation

**Coursework** Computing Systems (C, pointers, memory) · Discrete Mathematics · Linear Algebra | **Languages** English · Bengali · Hindi